Software Engineering II

Course Description

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Lecturers
Computer Engineering Program
School of Engineering
Universidad Nacional de Colombia

2026-I

Outline

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Overview

In the **Software Engineering 2** course, students will explore *advanced development methodologies*, including the definition of functional requirements, the design of basic architectures such as `MVC` and `hexagonal`, and the implementation of `APIs` and `NoSQL databases`.

Additionally, a DevOps culture will be encouraged through `continuous integration` and `deployment practices`. Through hands-on prototypes, students will apply these concepts to develop robust and scalable software systems.

Goals

The objective is to provide students with advanced training in the fundamentals of software development, equipping them with the *skills* needed to `design`, `implement`, and `maintain` modern software systems using agile methodologies, effective architectures, and DevOps practices.

Through *hands-on projects*, we aim to cultivate a deep understanding of modern tools and techniques in the field.

Learning Results I

As follows, the main learning results for this course:

- **Functional Requirements**: Define, refine, and manage requirements using user stories, acceptance criteria, and story points.
- **Software Architectures**: Design and implement basic architectures such as MVC and hexagonal architecture.
- **Domain-Driven Design**: Apply DDD principles to model complex domains.
- **Design Patterns**: Apply key design patterns — *creational*, *structural*, and *behavioral* — to improve software quality.
- **Web Development**: Develop web applications covering frontend, backend, Web APIs, and HTTP communication.

Learning Results II

As follows, the main learning results for this course:

- **APIs** and **HTTP**: Design and document REST APIs following best practices.
- **Software Testing**: Implement integration testing and test-driven development (TDD) strategies using tools such as `Pytest`, `Cucumber`, and `Apache JMeter`.
- **DevOps** and **CI/CD**: Explain the DevOps philosophy and apply CI/CD practices using tools like `LocalStack`, `GitHub Actions`, `Docker`, and `Kubernetes`.

Pre-Requisites

This is a basic course, so you must have some knowledge in:

- **Programming** in Java, Python, or C++.
- **Object-Oriented Programming** concepts like classes, objects, inheritance, and polymorphism.
- **Git** basic usage, and **GitHub** basic usage.
- Use of **IDEs** like VS Code, Eclipse, or PyCharm.
- Basic knowledge of **databases** and **SQL**.

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Syllabus I

Time	Topic
Week 1	Agile Fundamentals
Week 2-3	Refining and Prototyping: *User stories, acceptance criteria (Given-When-Then), Story Mapping, Estimation, Fast Prototyping*
Week 4	Domain-Driven Design: *Hexagonal Architectures*
Week 5-6	SOLID principles: *User Stories Relation, Good Practices, Quality Attributes*
Week 7-8	Patterns Design I: *Creational*
Week 8-9	Patterns Design II: *Structural*
Week 10-11	Patterns Design III: *Behavioral*

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Syllabus II

Time	Topic
Week 11-12	Web Applications: *Frontend*, *Backend*, *Web API*, *HTTP communication*
Week 13	Test Engineering: *Test-Driven Design (TDD)*, *Pytest*, *Cucumber*, *Apache JMeter*
Week 14	DevOps: *LocalStack*, *GitHub Actions (CI)*, *Docker*, *Kubernetes*, *Cloud Computing*

Grades Percentages

Item	Percentage
Quizzes	10%
Final Test	20%
Laboratories & Workshops	30%
Course Project	40%

30% Theor.

70% Pract.

Code of Conduct

- Always be **respectful** to your classmates and to me. You must be **kind** to everyone inside (*and outside*) the classroom.
- There is no best programming language, tool, or technology. There are only **better** or **worse** solutions.
- You must be **honest** with your work. If you don't know something, just **ask** us. We are glad to help you.
- You must be **responsible** with your work. If you don't submit **on time**, please don't complain.
- You must not be **disruptive** or **negatively affect** the **classroom environment**.

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Bibliography

Recommended bibliography:

- **Object-Oriented Analysis and Design**, by Grady Booch.
- **Design Patterns: Elements of Reusable Object-Oriented Software**, by Erich Gamma, Richard Helm, Ralph Johnson, & John Vlissides.
- **Clean Code: A Handbook of Agile Software Craftsmanship**, by Robert C. Martin.
- **The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations**, by Gene Kim, Jez Humble, Patrick Debois, & John Willis.
- **Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation**, by Jez Humble & David Farley.

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Thanks!

Questions?